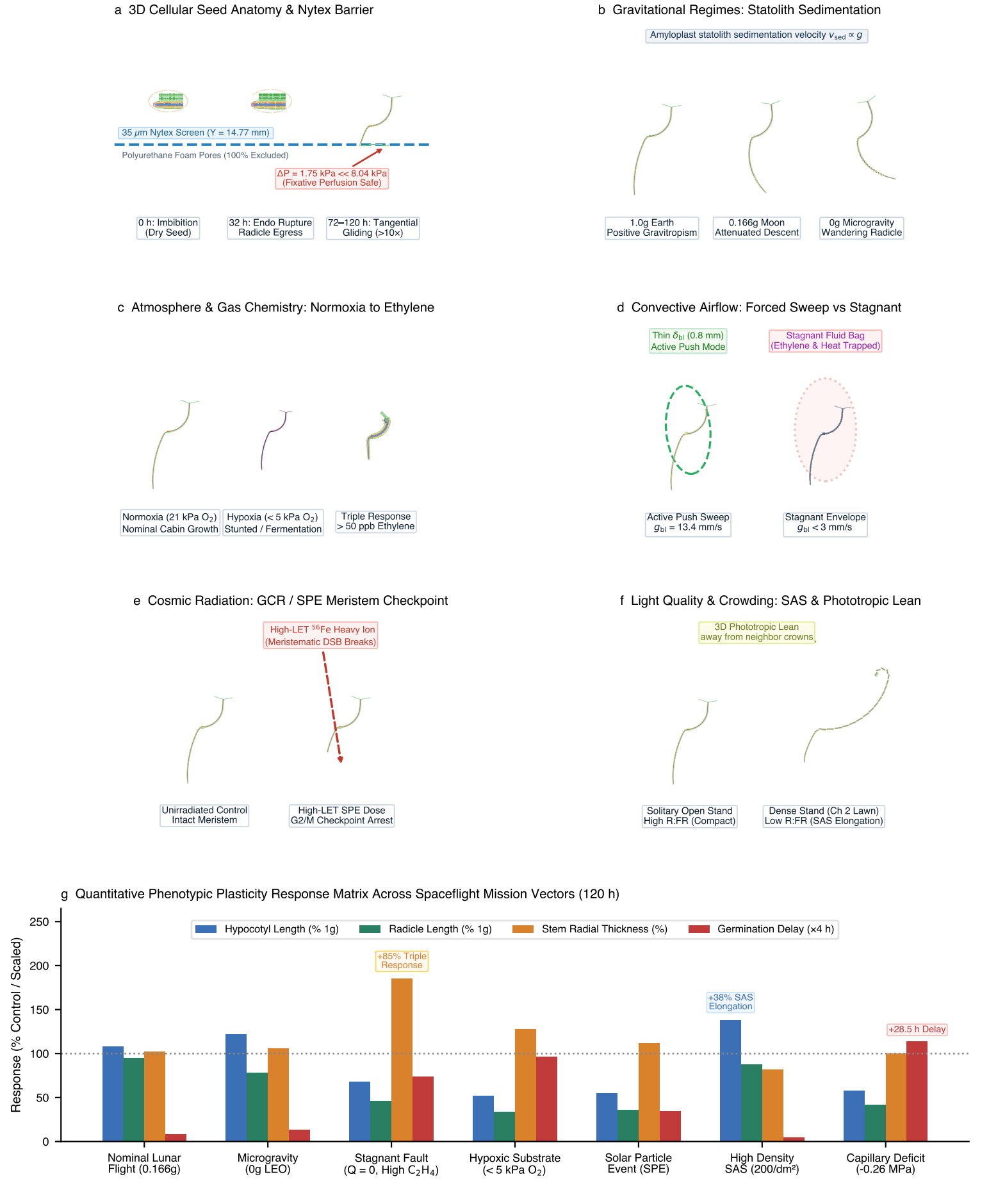

a 3D Cellular Seed Anatomy & Nytex Barrier
35 μm Nytex Screen (Y = 14.77 mm)
Polyurethane Foam Pores (100% Excluded)
ΔP = 1.75 kPa << 8.04 kPa
(Fixative Perfusion Safe)
0 h: Imbibition
(Dry Seed)
32 h: Endo Rupture
Radicle Egress
72–120 h: Tangential
Gliding (>10×)
b Gravitational Regimes: Statolith Sedimentation
Amyloplast statolith sedimentation velocity $v_{sed} \propto g$
1.0g Earth
Positive Gravitropism
0.166g Moon
Attenuated Descent
0g Microgravity
Wandering Radicle
c Atmosphere & Gas Chemistry: Normoxia to Ethylene
Normoxia (21 kPa O_2)
Nominal Cabin Growth
Hypoxia (< 5 kPa O_2)
Stunted / Fermentation
Triple Response
> 50 ppb Ethylene
d Convective Airflow: Forced Sweep vs Stagnant
Thin δ_{bl} (0.8 mm)
Active Push Mode
Stagnant Fluid Bag
(Ethylene & Heat Trapped)
Active Push Sweep
g_{bl} = 13.4 mm/s
Stagnant Envelope
g_{bl} < 3 mm/s
e Cosmic Radiation: GCR / SPE Meristem Checkpoint
High-LET ^{56}Fe Heavy Ion
(Meristematic DSB Breaks)
Unirradiated Control
Intact Meristem
High-LET SPE Dose
G2/M Checkpoint Arrest
f Light Quality & Crowding: SAS & Phototropic Lean
3D Phototropic Lean
away from neighbor crowns
Solitary Open Stand
High R:FR (Compact)
Dense Stand (Ch 2 Lawn)
Low R:FR (SAS Elongation)
g Quantitative Phenotypic Plasticity Response Matrix Across Spaceflight Mission Vectors (120 h)
Hypocotyl Length (% 1g)
Radicle Length (% 1g)
Stem Radial Thickness (%)
Germination Delay (×4 h)
Response (% Control / Scaled)
250
200
150
100
50
0
+85% Triple Response
+38% SAS Elongation
+28.5 h Delay
Nominal Lunar Flight (0.166g)
Microgravity (0g LEO)
Stagnant Fault (Q = 0, High C_2H_4)
Hypoxic Substrate (< 5 kPa O_2)
Solar Particle Event (SPE)
High Density SAS (200/dm²)
Capillary Deficit (-0.26 MPa)